Java Reserved Words

In Java, reserved words (also known as keywords) are predefined identifiers that have a specific meaning in the language and are used to define the structure and behavior of Java programs. These words cannot be used as identifiers (such as variable names, method names, class names, etc.) because they have a predefined function in the Java language syntax.

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1. Access Modifiers

These keywords control the visibility and access level of classes, methods, and variables.

- `public` : Makes a class, method, or variable accessible from anywhere in the program.
- `private` : Limits access to the class or method to within the same class.
- `protected` : Allows access within the same package and by subclasses.
- `default` (Package-Private): If no access modifier is specified, the member is accessible only within the same package.

2. Data Types

These keywords represent the primitive data types or refer to other data types in Java.

- `byte` : Represents an 8-bit signed integer.
- `short` : Represents a 16-bit signed integer.
- `int` : Represents a 32-bit signed integer.
- `long` : Represents a 64-bit signed integer.
- `float` : Represents a 32-bit floating-point number.
- `double` : Represents a 64-bit floating-point number.
- `char` : Represents a single 16-bit Unicode character.
- `boolean` : Represents a boolean value (either `true` or `false`).

3. Control Flow Statements

These reserved words are used to control the flow of execution in Java programs.

- `if` : Used for conditional execution.
- `else` : Specifies the alternative block of code if the `if` condition is false.
- `switch` : Used to specify multiple possible execution paths based on the value of an expression.
- `case` : Defines a branch in a `switch` statement.

- `default`: Specifies the default block in a `switch` statement when no `case` matches.
- `while`: Used to create a loop that repeats as long as a condition is true.
- `do`: Used to create a loop that executes at least once before checking the condition.
- `for`: Used to create a loop that executes a fixed number of times.
- `break`: Terminates the current loop or switch statement.
- `continue`: Skips the current iteration of the loop and moves to the next one.
- `return`: Exits from a method and optionally returns a value.

4. Exception Handling

These keywords are used to handle exceptions in Java.

- `try`: Defines a block of code that might throw an exception.
- `catch`: Defines a block of code that handles exceptions thrown in the `try` block.
- `finally`: Defines a block of code that is always executed after the `try` and `catch` blocks, regardless of whether an exception was thrown.
- `throw`: Used to explicitly throw an exception.
- `throws`: Declares the exceptions that a method may throw.

5. Class and Object Creation

These reserved words define classes, interfaces, and how objects are created.

- `class`: Defines a class.
- `interface`: Defines an interface.
- `extends`: Indicates that a class is inheriting properties from another class (for inheritance).
- `implements`: Indicates that a class is implementing an interface.
- `new`: Creates new objects or arrays.
- `this`: Refers to the current instance of the class.

6. Object-Oriented Keywords

These keywords are used to define object-oriented features like inheritance and polymorphism.

- `super`: Refers to the superclass of the current object.
- `abstract`: Defines an abstract class or method that must be implemented by subclasses.
- `final`: Used to define constants, methods that cannot be overridden, or classes that cannot be subclassed.
- `static`: Defines a class-level variable or method, accessible without instantiating the class.

- `synchronized`: Marks a method or block of code as synchronized (for thread safety).
- `volatile`: Indicates that a variable's value may be changed by multiple threads.

7. Package and Import

These keywords are used to define packages and import other classes or entire packages.

- `package`: Defines the package that a class belongs to.
- `import`: Imports a class or a package for use in the current file.

8. Other Keywords

These reserved words are used for various purposes such as typecasting, defining native methods, etc.

- `void`: Specifies that a method does not return any value.
- `instanceof`: Tests whether an object is an instance of a particular class or subclass.
- `native`: Specifies that a method is implemented in native code (e.g., C or C++).
- `strictfp`: Ensures consistent floating-point calculations across platforms (rarely used).
- `transient`: Prevents a variable from being serialized.
- `instanceof`: Tests if an object is an instance of a specified class or interface.

9. Package/Module System (Introduced in Java 9)

These keywords relate to the module system introduced in Java 9.

- `module`: Defines a module in the module system.
- `requires`: Specifies dependencies between modules.
- `exports`: Specifies which packages are accessible from other modules.
- `opens`: Allows deep reflection into a package at runtime.
- `provides`: Specifies a service implementation within a module.
- `uses`: Declares a service that the module uses.
